

Detailed derivations for human review

Companion to the finite-response RH formulation

Review record not yet completed

13 September 2026

Abstract

This file expands the chain of arguments needed for the condensed paper. It distinguishes the classical Weil criterion, unconditional operator calculations, and the unproved arithmetic matrix conjecture. It also gives finite-input error bounds suitable for rigorous computation. The source investigation is repository revision [efb908a](#); this companion is a new exposition, not an independently refereed validation of that manuscript. The human review should cover the hypotheses, domains, constants, quantifiers and citations as well as the displayed algebra. No verification of RH is claimed.

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1 What must be reviewed

The proposed short paper does not use the interacting vacuum metrics, physical residue normalization, the conditional tt^* equation, the remote delay construction, or the graph realization as premises. Those results remain in the longer investigation. The short paper needs only the following chain:

$$\begin{array}{c}
\text{Weil criterion and its exact arithmetic normalization} \\
\Downarrow \\
\text{closed logarithmic form and explicit boundary correction} \\
\Downarrow \\
\text{unconditional high-mode positivity and well-defined mixed block} \\
\Downarrow \\
\text{Schur identity on the full form domain} \\
\Downarrow \\
\text{RH} \iff S_n \succeq 0 \text{ for every } n \geq 1.
\end{array}$$

The remainder estimates provide a constructive way to approximate S_n ; they do not prove its sign. The positive shifted reference supplies an equivalent contraction statement but is not needed for the Schur identity.

Throughout, $X_L = L^2(0, L)$ over $\mathbb{C}$, inner products are antilinear first, $F = E_L f$ is the zero extension, and $\hat{F}(\tau) = \int F(x) e^{-i\tau x} dx$. The full-line shift is $U_d F(x) = F(x - d)$. Set $a_k = 2k + 1/2$ and write b for the scalar gamma multiplier, reserving B for the mixed operator. These differ from some letter choices in the long manuscript.

2 The arithmetic normalization, including the contact

For $g = F * \tilde{F}$, $\tilde{F}(x) = \overline{F(-x)}$, one has

$$g(0) = \|F\|^2, \quad g(-r) = \overline{g(r)}, \quad \text{Re } g(r) = \text{Re} \langle F, U_r F \rangle =: h_f(r).$$

With $n_\gamma(r) = e^{-r/2}/(1 - e^{-2r})$ for $r > 0$, the additive Weil functional, in the normalization of [1, Section 1.1], is

$$\begin{aligned}
\mathcal{W}(g) &= \int_{\mathbb{R}} 2 \cosh(x/2) g(x) dx - \sum_{p,m} (\log p) p^{-m/2} \{g(m \log p) + g(-m \log p)\} \\
&\quad - (\log(4\pi) + \gamma_E) g(0) - \int_0^\infty [g(r) + g(-r) - 2e^{-r/2} g(0)] n_\gamma(r) dr.
\end{aligned} \tag{2.1}$$

This equality is initially used on smooth compact support. The published criterion states that RH is equivalent to $\mathcal{W}(F * \tilde{F}) \geq 0$ for every complex $F \in C_c^\infty(\mathbb{R})$. This is the one external number-theoretic equivalence on which the short paper rests.

The gamma energy is

$$\begin{aligned}
K[F] &= \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \\
&= \int_0^\infty [2g(0) - g(r) - g(-r)] n_\gamma(r) dr.
\end{aligned} \tag{2.2}$$

Consequently the last line of (2.1) equals $K[F] + w_0\|F\|^2$, where

$$w_0 = -\log(4\pi) - \gamma_E - 2 \int_0^\infty (1 - e^{-r/2}) n_\gamma(r) dr.$$

The integral is convergent at zero and infinity. Expanding the denominator and integrating positive terms gives

$$\begin{aligned} \int_0^\infty (1 - e^{-r/2}) n_\gamma(r) dr &= \sum_{k \geq 0} \left(\frac{1}{2k + 1/2} - \frac{1}{2k + 1} \right) \\ &= \frac{1}{2} [\psi(1/2) - \psi(1/4)]. \end{aligned}$$

Using $\psi(1/2) = -\gamma_E - 2 \log 2$ proves

$$w_0 = \psi(1/4) - \log \pi = -\gamma_E - \pi/2 - 3 \log 2 - \log \pi. \quad (2.3)$$

This checks the negative contact without assigning it by a normalization of a positive state.

For the poles, convolution gives the kernel $2 \cosh((x - y)/2)$. On $(0, L)$, with $c(x) = \cosh((x - L/2)/2)$ and $s(x) = \sinh((x - L/2)/2)$, the identity $\cosh(u - v) = \cosh u \cosh v - \sinh u \sinh v$ gives $2cc^* - 2ss^*$. In particular the pole term has one positive and one negative direction, despite its pointwise positive kernel. The prime correlations vanish for $m \log p \geq L$. Thus the precise form is

$$Q_L[f] = K[E_L f] + w_0\|f\|^2 + 2|\langle c, f \rangle|^2 - 2|\langle s, f \rangle|^2 - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \quad (2.4)$$

3 Gamma expansion, source domain and cores

The digamma partial fractions [3] give

$$\psi(z + it) - \psi(z) = \sum_{k \geq 0} \left[\frac{1}{k + z} - \frac{1}{k + z + it} \right].$$

At $z = 1/4$, taking real parts and substituting $t = \sqrt{s}/2$ yields

$$b(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{a_k^2 + s}. \quad (3.1)$$

To identify this multiplier with the spatial energy in Section 2, use $2g(0) - g(r) - g(-r) = \|F - U_r F\|^2$ and Plancherel:

$$\begin{aligned} K[F] &= \frac{1}{2\pi} \int_{\mathbb{R}} \left[2 \int_0^\infty (1 - \cos(\tau r)) n_\gamma(r) dr \right] |\widehat{F}(\tau)|^2 d\tau, \\ 2 \int_0^\infty (1 - \cos(\tau r)) e^{-ar} dr &= \frac{2}{a} - \frac{2a}{a^2 + \tau^2} = \frac{2}{a} \frac{\tau^2}{a^2 + \tau^2}. \end{aligned} \quad (3.2)$$

Since $n_\gamma(r) = \sum_{k \geq 0} e^{-a_k r}$ and the integrands are nonnegative, Tonelli justifies both interchanges, including infinite energy for arbitrary $F \in L^2(\mathbb{R})$. Thus $K[E_L f] = t_L[f]$. This explicitly links the cited Weil normalization to the operator used below.

The series converges locally uniformly in $s \geq 0$. Its summands are nonnegative and increasing in s , and at least one is strictly increasing. Digamma asymptotics give $b(\tau^2) = \log |\tau| - \log 2 - \psi(1/4) + O(\tau^{-2})$. Thus $1 + b(\tau^2)$ is comparable above and below to $1 + \log(2 + |\tau|)$.

For $a > 0$, minimizing

$$\mathcal{E}_a(F, u) = \frac{2}{a}(\|F - u\|^2 + a^{-2}\|u'\|^2)$$

gives $\hat{u} = a^2 \hat{F}/(a^2 + \tau^2)$ and minimum $\int (2/a)\tau^2/(a^2 + \tau^2)|\hat{F}|^2 d\tau/(2\pi)$. This can alternatively be represented by the output column

$$\left(\sqrt{2/a} H_{\mathbb{R}}(a^2 + H_{\mathbb{R}})^{-1} F, \sqrt{2a} \partial_x (a^2 + H_{\mathbb{R}})^{-1} F \right), \quad H_{\mathbb{R}} = -\partial_x^2.$$

Squaring the two components gives exactly the same multiplier. The orthogonal sum of these columns is a closed source: convergence of input and output identifies every bounded component, and the square-summability of the limiting output is exactly membership in its maximal domain. That domain after zero extension is

$$\mathcal{D}_L = \{f : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty\}. \quad (3.3)$$

The resulting form t_L is closed. Every prime shift is a contraction after compression, and the two pole functions are in $L^2(0, L)$, so their finite sum R_L is a bounded self-adjoint operator. Bounded perturbation gives a closed semibounded form $t_L + R_L$ and the operator $W_L = T_L + R_L$ on $\text{Dom } T_L$.

3.1 Smooth compact support is a form core

Translate to $(-L/2, L/2)$ for this proof. Given the zero extension F , let $D_r F(x) = r^{-1/2} F(x/r)$, $r < 1$. The Fourier transform is $r^{1/2} \hat{F}(r\tau)$. Changing variable $\sigma = r\tau$ shows that the weighted norm uses $1 + \log(2 + |\sigma|/r)$. For r in, say, $[1/2, 1]$, this is uniformly comparable to the original weight. Dilations converge strongly as $r \uparrow 1$: first prove it on Schwartz functions by dominated convergence, then use their density in the weighted Fourier space and uniform boundedness. The support of $D_r F$ lies strictly inside the interval. Convolution with a smooth approximate identity of radius smaller than the support gap stays inside the interval, and its bounded Fourier multiplier converges to one in the same weighted norm. This proves the required core assertion. No pointwise trace at the interval endpoints is imposed on $\mathcal{D}_L$.

3.2 Compactness and the scalar truncation error

For a bounded set in the form norm, its Fourier mass outside $[-R, R]$ is bounded by a constant divided by $\log(2 + R)$. The map from a fixed Fourier band to $L^2(0, L)$ has a square-integrable kernel and is compact. Approximating the embedding by these compact maps shows that $\mathcal{D}_L \hookrightarrow X_L$ is compact. The associated operators therefore have compact resolvent. They can have finitely many negative eigenvalues; this is not their absence.

For numerical enclosure of $b(s)$, positivity and an integral test give

$$0 \leq b(s) - \sum_{k < J} \frac{2}{a_k} \frac{s}{a_k^2 + s} \leq s \left(\frac{2}{a_J^3} + \frac{1}{2a_J^2} \right). \quad (3.4)$$

Indeed $a_k = a_J + 2(k - J)$, and $\int_J^\infty 2(2x + 1/2)^{-3} dx = 1/(2a_J^2)$.

4 Neumann–Dirichlet comparison and the cutoff

For one mass, restricting a whole-line competitor u to $(0, L)$ drops the nonnegative exterior energy. Minimizing over all $H^1(0, L)$ competitors yields the Neumann resolvent. In the other direction, a Dirichlet competitor extends by zero to an admissible whole-line competitor. Thus, for every finite tower,

$$b_J(H_N) \leq T_{L,J} \leq b_J(H_D).$$

Monotone limits of the nonnegative forms preserve this order and its domain inclusions. In particular

$$b(H_N) \leq T_L \leq b(H_D). \quad (4.1)$$

Min–max, indexing eigenvalues from one, gives

$$b(((j-1)\pi/L)^2) \leq \lambda_j(T_L) \leq b((j\pi/L)^2).$$

The logarithmic asymptotic then yields $\lambda_j(T_L) = \log(\pi j/(2L)) - \psi(1/4) + O_L(j^{-1})$.

For one prime p , put $d = \log p$ and $q = p^{-1/2}$. Express $x = t + kd$ with $0 < t < d$ and retain those integers k for which $0 < x < L$. This decomposes X_L into a direct integral of finite chain spaces. Their lengths are at most $h = \lceil L/d \rceil$. The complete off-diagonal correlation on a chain is the matrix with entries $q^{|i-j|}$ for $i \neq j$, zero on the diagonal. Its operator norm is bounded by the maximum absolute row sum $r_h(q)$. The row i has sum $\sigma_i = \sum_{j=1}^{i-1} q^j + \sum_{j=1}^{h-i} q^j$. Since $\sigma_{i+1} - \sigma_i = q^i - q^{h-i}$, the maximum is attained at the central row or rows. Summing the two geometric series there gives

$$r_h(q) = \frac{q}{1-q} \left(2 - q^{\lfloor (h-1)/2 \rfloor} - q^{\lceil (h-1)/2 \rceil} \right).$$

This also gives $r_1 = 0$. Therefore the negative prime contribution is bounded below by $-dr_h(q)I$. The assertion also holds when L/d is an integer; endpoint chains of exceptional length occur only on null sets.

Direct integration gives

$$\|c\|^2 = L/2 + \sinh(L/2), \quad \|s\|^2 = -L/2 + \sinh(L/2), \quad \langle c, s \rangle = 0.$$

Dropping $2cc^*$ and bounding $-2ss^*$ proves

$$W_L \geq b(H_N) - \beta_L I, \quad \beta_L = -w_0 + 2 \sinh(L/2) - L + \sum_{\log p < L} (\log p) r_{\lceil L/\log p \rceil}(p^{-1/2}). \quad (4.2)$$

An optional coarser prime term is $2 \sum_{m \log p < L} (\log p) p^{-m/2}$.

4.1 A certified cutoff without an equality decision

Put $P_h = \sum_{j=0}^{h-1} e_j e_j^*$ and $\delta_0 = 1/16$. For any $L > 0$, choose an integer $N(L)$ with $b_{N(L)} > \beta_L + \delta_0$, and set $P = P_{N(L)}$, $Q = I - P$. Such integers exist since $b_j \rightarrow \infty$. For integer lengths $L = n$, fix deterministic rational enclosure routines for β_n and b_j . The following convention then specifies $N(n)$. At stage $r = 1, 2, \dots$, compute intervals

$$\beta_n \in [\underline{\beta}_{n,r}, \bar{\beta}_{n,r}], \quad b_j \in [\underline{b}_{j,r}, \bar{b}_{j,r}] \quad (1 \leq j \leq r),$$

each of width at most 2^{-r} . If any $\underline{b}_{j,r} > \bar{\beta}_{n,r} + \delta_0$, stop and choose the least such j at that stage. All comparisons at this step are rational.

For termination, choose a fixed j_* with $\epsilon = b_{j_*} - \beta_n - \delta_0 > 0$. At any stage $r \geq j_*$ with $2^{1-r} < \epsilon$, one has

$$\underline{b}_{j_*,r} - \bar{\beta}_{n,r} - \delta_0 \geq \epsilon - 2^{1-r} > 0.$$

The procedure therefore stops. An equality at a smaller index cannot block it, because the search increases both the candidate range and the accuracy, rather than waiting for that single comparison.

Here is why the needed enclosure routines exist without using RH. Enumerate primes and their powers below the integer $3^n > e^n$ and refine rational bounds for e^n to decide which lie below e^n . Every comparison eventually resolves: e^n is transcendental, whereas each prime power is an integer. The same comparisons determine $\lceil n/\log p \rceil$, using $p^{h-1} < e^n < p^h$ for active primes. This deals with the discontinuous integer choices in β_n . The remaining expressions are finite continuous arithmetic combinations of computable constants. One can use Taylor series with remainder bounds, the atanh series for logarithms, bisection for square roots, and the alternating series in $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. For Euler's constant the intervals $[H_M - \log(M+1), H_M - \log M]$ shrink to γ_E as $M \rightarrow \infty$. Finally (3.4), monotonicity in s , and rational bounds for π enclose each b_j . Fixing these routines, truncation rules and refinement order in advance makes the search deterministic. The theorem holds for every such fixed implementation; a computation should record its routine and the cutoff certificate.

The original mathematical convention $N_{\min}(L) = \min\{j \geq 1 : b_j \geq \beta_L + \delta_0\}$ remains valid, but equality need not be decidable from convergent enclosures. No algorithm to identify that exact minimum is asserted. The certified convention can return a different dimension. Each admissible cutoff gives the same positivity criterion, by the Schur identity in Section 7. In particular the explicit $N = 4$ certificate in Section 13 may be used as a separate convention at length one.

The operator H associated with the high form on $Q\mathcal{D}_L$ has the lower bound

$$H \geq \Lambda, \quad \Lambda e_j = d_j e_j, \quad d_j = b((j\pi/L)^2) - \beta_L \geq \delta_0 \quad (j \geq N).$$

Its inverse is legitimate before RH is assumed. The comparison uses the gamma operator and a crude bound on the signed remainder; it does not select low modes by assuming the spectrum of W_L is positive.

5 Deriving and controlling the exact boundary correction

For a mass $a > 0$, let $R_{f,a}$ be the compressed free resolvent and $R_{N,a}$ the Neumann resolvent. Their kernels are

$$R_{f,a}(x, y) = \frac{e^{-a|x-y|}}{2a}, \quad R_{N,a}(x, y) = \frac{\cosh(a \min(x, y)) \cosh(a(L - \max(x, y)))}{a \sinh(aL)}.$$

Both solve $(-\partial_x^2 + a^2)R = \delta_y$ in the open interval; the derivative jump is -1 . The second kernel has zero normal derivative at both endpoints. Expand its hyperbolic factors separately on $x \leq y$ and $x \geq y$. Subtraction gives

$$2a(R_{N,a} - R_{f,a}) = K_a = \frac{u_a u_a^* + v_a v_a^* + r_a(u_a v_a^* + v_a u_a^*)}{1 - r_a^2}, \quad (5.1)$$

where $u_a = e^{-ax}$, $v_a = e^{-a(L-x)}$, $r_a = e^{-aL}$. The coefficient matrix is $(1 - r_a^2)^{-1} \begin{pmatrix} 1 & r_a \\ r_a & 1 \end{pmatrix}$, with positive eigenvalues $1/(1 - r_a)$ and $1/(1 + r_a)$. The energy summand is $(2/a)I - 2aR$, so the free compressed energy is the Neumann energy plus K_a , with the positive sign.

5.1 Uniform boundedness of the infinite sum

Let $n(t) = \sum_{k \geq 0} e^{-a_k t} = e^{-t/2}/(1 - e^{-2t})$. In the kernel of K_a , the first $u_a u_a^*$ and $v_a v_a^*$ image terms sum to $n(x+y) + n(2L-x-y)$. Every additional diagonal image has an extra factor e^{-2aL} ; the mixed images have path length at least L . Thus the remainder is bounded, for example by

$$\frac{2[n(2L) + n(L)]}{1 - e^{-L}}.$$

As $n(t) - (2t)^{-1}$ is bounded on $(0, 2L]$, the sum is

$$K_L = \frac{1}{2}(C_0 + C_L) + \text{a bounded-kernel operator}, \quad (5.2)$$

where $C_0(x, y) = (x + y)^{-1}$ and $C_L(x, y) = (2L - x - y)^{-1}$. Using weight $x^{-1/2}$, the identity

$$\int_0^\infty \frac{y^{-1/2}}{x + y} dy = \pi x^{-1/2}$$

proves $\|C_0\| \leq \pi$ by the weighted Schur test; reflection treats C_L . The same kernel majorant bounds every positive partial sum. The monotone bounded sequence $K^{(J)} = \sum_{k < J} K_{a_k}$ converges strongly to a bounded positive K_L .

For any $f \in X_L$, finite-tower energies satisfy $t_{L,J}[f] = \langle f, b_J(H_N)f \rangle + \langle f, K^{(J)}f \rangle$. The last term has a bounded limit for every f . Hence the two increasing energy limits are finite for exactly the same f , and their closed forms differ by K_L . This proves

$$T_L = b(H_N) + K_L, \quad \text{Dom } T_L = \text{Dom } b(H_N), \quad \mathcal{D}_L = \text{Dom } b(H_N)^{1/2}. \quad (5.3)$$

Spectral partial sums of $b(H_N)$ are an operator core; bounded perturbation preserves that fact. In particular all finite cosine sums belong to $\text{Dom } W_L$, not merely to its form domain.

5.2 Why the boundary operator is not compact

Split $(0, L)$ at $L/2$. In (5.2), all off-diagonal blocks and the opposite endpoint kernel within each diagonal block are bounded kernels. They are Hilbert–Schmidt. Modulo compact operators, K_L is the direct sum of two half-Carleman operators. Thus its essential norm is at most $\pi/2$.

For the reverse bound take $\phi_R(t) = t^{-1/2} 1_{[1, R]}(t)/\sqrt{\log R}$ on $(0, \infty)$. Changing to $t = e^u$ transforms the Carleman kernel to convolution by $k(u - v) = [2 \cosh((u - v)/2)]^{-1}$. Hence its Rayleigh quotient is

$$\int_{-\log R}^{\log R} \left(1 - \frac{|v|}{\log R}\right) k(v) dv \longrightarrow \int_{\mathbb{R}} k(v) dv = \pi.$$

For fixed R , dilate into $(\epsilon, \epsilon R) \subset (0, L/2)$. The unit vectors tend weakly to zero and keep their Carleman quotient, while every compact quadratic term tends to zero. First let $\epsilon \downarrow 0$, then $R \rightarrow \infty$. This gives the reverse essential-norm bound and proves

$$\|K_L\|_{\text{ess}} = \pi/2, \quad \|K_L - K^{(J)}\| \geq \pi/2 \quad \text{for every finite } J.$$

This does not conflict with compact resolvent of the unbounded T_L . The needed Carleman facts are proved here; [5] gives broader context.

6 Matrix entries and finite-input tails

Let $\omega_j = j\pi/L$ and $e_j = \nu_j \cos(\omega_j x)$. Integration gives

$$\langle u_a, e_j \rangle = \frac{\nu_j a (1 - (-1)^j e^{-aL})}{a^2 + \omega_j^2}, \quad \langle v_a, e_j \rangle = (-1)^j \langle u_a, e_j \rangle.$$

Inserting this into (5.1) cancels the denominator and yields

$$K_a e_j = \frac{\nu_j a}{a^2 + \omega_j^2} (u_a + (-1)^j v_a). \quad (6.1)$$

Taking a second inner product gives zero for opposite parities, and for equal parities gives

$$\langle e_i, K_L e_j \rangle = 2\nu_i \nu_j \sum_{k \geq 0} \frac{a_k^2 (1 - (-1)^j e^{-a_k L})}{(a_k^2 + \omega_i^2)(a_k^2 + \omega_j^2)}. \quad (6.2)$$

6.1 A faster tail for finite compressions

For equal-parity entries the rational part of each summand in (6.2) is at most a_k^{-2} , and $|1 - (-1)^j e^{-a_k L}| \leq 1 + e^{-a_J L}$ for $k \geq J$. Hence

$$|\langle e_i, (K_L - K^{(J)}) e_j \rangle| \leq 2\nu_i \nu_j (1 + e^{-a_J L}) \left(a_J^{-2} + \frac{1}{2a_J} \right). \quad (6.3)$$

The integral test used here is $\sum_{k \geq J} a_k^{-2} \leq a_J^{-2} + (2a_J)^{-1}$. Opposite-parity entries are exactly zero. For a unit coefficient vector, bound the absolute quadratic form by the rank-one majorant with entries given by the right hand side of (6.3), and use $\sum_{j < h} \nu_j^2 = (2h - 1)/L$. This proves

$$\|P_h (K_L - K^{(J)}) P_h\| \leq \frac{2(2h - 1)}{L} (1 + e^{-a_J L}) \left(a_J^{-2} + \frac{1}{2a_J} \right). \quad (6.4)$$

This is $O(J^{-1})$ for fixed L, h and is useful for low and finite Galerkin matrices. It does not control the entire output column.

6.2 The full-column tail

Since $\|u_a\|, \|v_a\| \leq (2a)^{-1/2}$, equation (6.1) gives $\|K_a e_j\| \leq \sqrt{2} \nu_j a^{-3/2}$. The integral test yields $\sum_{k \geq J} a_k^{-3/2} \leq a_J^{-3/2} + a_J^{-1/2}$. The operator norm on the first h columns is at most their Hilbert–Schmidt norm. Using $\sum_{j < h} \nu_j^2 = (2h - 1)/L$ proves

$$\|(K_L - K^{(J)}) P_h\| \leq \sqrt{\frac{2(2h - 1)}{L}} (a_J^{-1/2} + a_J^{-3/2}) =: \eta_{h,J}. \quad (6.5)$$

Only the input dimension h is fixed in this limit. It does not imply operator-norm convergence on the full space.

6.3 Prime and pole entries without an unspecified kernel

Define

$$\mathcal{I}(\lambda, \phi; a, b) = \begin{cases} [\sin(\lambda b + \phi) - \sin(\lambda a + \phi)]/\lambda, & \lambda \neq 0, \\ (b - a) \cos \phi, & \lambda = 0. \end{cases}$$

For $0 < d < L$, product-to-sum gives

$$t_{ij}(d) := \langle e_i, V_d e_j \rangle = \frac{\nu_i \nu_j}{2} [\mathcal{I}(\omega_i - \omega_j, \omega_j d; d, L) + \mathcal{I}(\omega_i + \omega_j, -\omega_j d; d, L)]. \quad (6.6)$$

The adjoint entry is $t_{ji}(d)$. This handles the zero-frequency cases by the second branch of $\mathcal{I}$.

For $\sigma \neq 0$ set

$$J_j(\sigma) = \int_0^L e^{\sigma x} \cos(\omega_j x) dx = \frac{\sigma[(-1)^j e^{\sigma L} - 1]}{\sigma^2 + \omega_j^2}.$$

Then

$$\begin{aligned} c_j &:= \langle e_j, c \rangle = \frac{\nu_j}{2} [e^{-L/4} J_j(1/2) + e^{L/4} J_j(-1/2)], \\ s_j &:= \langle e_j, s \rangle = \frac{\nu_j}{2} [e^{-L/4} J_j(1/2) - e^{L/4} J_j(-1/2)]. \end{aligned}$$

Thus the entire cosine matrix is

$$w_{ij} = (b_j + w_0)\delta_{ij} + k_{ij} + 2c_i c_j - 2s_i s_j - \sum_{m \log p < L} (\log p) p^{-m/2} [t_{ij}(m \log p) + t_{ji}(m \log p)]. \quad (6.7)$$

All entries are real, but the quadratic-form assertions apply to complex inputs. For residual norms, the full columns are also explicit:

$$\begin{aligned} W_L e_j &= (b_j + w_0)e_j + K_L e_j + 2c_j c - 2s_j s \\ &\quad - \sum_{d=m \log p < L} (\log p) p^{-m/2} [1_{x>d} e_j(x-d) + 1_{x<L-d} e_j(x+d)]. \end{aligned} \quad (6.8)$$

Finite mass sums in (6.1) turn these into finite sums of exponentials, trigonometric functions, and their interval restrictions. Products and squared norms are elementary integrals on the finitely many overlap intervals. Equations (3.4) and (6.5) supply the missing infinite tails. Use (6.4) for finite compressions and (6.5) for full residual functions.

6.4 Reflection of the complete arithmetic operator

Let $\mathcal{J}f(x) = f(L - x)$. This is a self-adjoint unitary, $\mathcal{J}e_j = (-1)^j e_j$, and direct substitution gives

$$\mathcal{J}V_d \mathcal{J} = V_d^*, \quad \mathcal{J}c = c, \quad \mathcal{J}s = -s.$$

The gamma form is invariant under reflection since its Fourier multiplier is even. Equivalently both $b(H_N)$ and K_L commute with $\mathcal{J}$, the latter exchanging its two endpoint kernels. The pole and symmetrized prime terms also commute with $\mathcal{J}$. Thus the entire closed form and W_L commute with reflection, and so do the cosine projections P and Q .

On the even subspace, the negative pole term vanishes. One may therefore use the sharper remainder bound $\beta_L^{\text{even}} = -w_0 + \sum_{\log p < L} (\log p) r_{\lceil L/\log p \rceil}(p^{-1/2})$ there. The family in the short paper keeps the common cutoff for both parities; no separate choice is needed for its equivalence.

7 The finite response and both directions of the equivalence

Because $PX_L \subset \text{Dom } W_L$, both $A = PW_LP$ and $B = QW_LP$ are bounded on the finite low space. Restriction of the closed form defines $H \geq \delta_0 I$ on QX_L . Its operator domain is $Q \text{Dom } b(H_N)$, by (5.3), and on that domain $Hh = QW_L h$. This is a form compression; $W_L h$ may also have a low component. For $p = Pf$ and $h = Qf$ the form is

$$Q_L[p + h] = \|H^{1/2}h\|^2 + 2\text{Re}\langle h, Bp \rangle + \langle p, Ap \rangle.$$

As $H^{-1}Bp \in \text{Dom } H$, expand the square to obtain

$$Q_L[p + h] = \|H^{1/2}(h + H^{-1}Bp)\|^2 + \langle p, S_L p \rangle, \quad S_L = A - B^*H^{-1}B. \quad (7.1)$$

The expansion uses $H^{1/2}H^{-1} = H^{-1/2}$ and the closed form pairing. It does not require every h to be in $\text{Dom } H$.

If $S_L \succeq 0$, equation (7.1) gives $Q_L \geq 0$ on its full form domain. Conversely choose $h = -H^{-1}Bp$. The high square vanishes and $Q_L[p + h] = \langle p, S_L p \rangle$, so positivity of Q_L forces positivity of S_L . A negative vector of S_L therefore produces a negative form-domain input. Approximation by the smooth core produces a smooth negative input.

Under RH, the classical Weil criterion gives positivity on the smooth core; continuity in the shifted form norm extends it to $\mathcal{D}_L$. Thus $S_n \succeq 0$ for every integer $n \geq 1$. Conversely, every compactly supported test function can be translated into $(0, n)$ for an integer n . Translation changes its Fourier transform by a phase and preserves every translated correlation. The pole kernel $2 \cosh((x - y)/2)$ also depends only on the difference, so the entire form is invariant. Zero extension to a larger interval preserves the same form and merely adds inactive translations with zero overlap. Positivity of every S_n now implies the global Weil criterion, hence RH.

This argument uses only a cofinal sequence of supports. It requires neither source compatibility nor a uniform full-form gap. It also shows that an arbitrary finite list of successful lengths cannot finish the proof.

Reflection leaves the high form invariant, so its restricted reflection commutes with H and H^{-1} . The mixed map B intertwines the low and high reflections. Hence S_L commutes with the low reflection and

$$S_L = S_L^{\text{even}} \oplus S_L^{\text{odd}}, \quad \dim S_L^{\text{even}} = \lceil N/2 \rceil, \quad \dim S_L^{\text{odd}} = \lfloor N/2 \rfloor.$$

The finite Galerkin spaces preserve parity as well. Galerkin trials, or arbitrary trials chosen to intertwine reflection, give the same block decomposition of the enclosures. Both blocks are required for the stated fixed-interval criterion; no odd-only variant of Weil's theorem is used.

8 The positive matrix comparison and the source interpretation

For $\alpha = \beta_L + 1$, the operator $T = W_L + \alpha I \geq I$ is positive independently of RH. Its block matrix is

$$T = \begin{pmatrix} A + \alpha I & B^* \\ B & H + \alpha I \end{pmatrix}.$$

Let $V = (H + \alpha I)^{-1}$. Completing the shifted high square yields

$$G = A + \alpha I - B^*VB \geq I.$$

To check the bound, minimize $\langle p + h, T(p + h) \rangle \geq \|p\|^2 + \|h\|^2$ over h at fixed p . Define

$$D = \alpha I + B^*(H^{-1} - V)B = \alpha[I + B^*VH^{-1}B] \geq \alpha I.$$

The product VH^{-1} is positive because both factors are functions of the same H ; arbitrary products of positive operators would not justify this step. Direct subtraction gives $G - D = S_L$. Congruence by $G^{-1/2}$ therefore proves

$$S_L \succeq 0 \iff G^{-1/2}DG^{-1/2} \preceq I. \quad (8.1)$$

Both G and D can change with the permissible positive shift; their difference cannot. This rules out tuning a source normalization to improve the arithmetic sign.

For a precise source interpretation, take any closed densely defined $\mathcal{A} : X_L \rightarrow \mathcal{Y}$ with $\mathcal{A}^*\mathcal{A} = T$. Such a source exists, for example $\mathcal{A} = T^{1/2}$, and its domain is the form domain of T . Define the adjusted low output

$$Cp = \mathcal{A}(p - VBp), \quad p \in PX_L, \quad V = (H + \alpha I)^{-1}. \quad (8.2)$$

For high form-domain h , the T -form pairing gives

$$\langle \mathcal{A}h, Cp \rangle = \langle h, Bp \rangle - \langle h, (H + \alpha I)VBp \rangle = 0.$$

Thus Cp is the component of $\mathcal{A}p$ orthogonal to the closure of the high output range; the removed vector $\mathcal{A}VBp$ is in that range. The raw output $\mathcal{A}p$ need not be orthogonal to it. The shifted square identity gives $C^*C = G$. Take the abstract defect output to be $D^{1/2}p$, with Gram D .

Since $G \geq I$, C is injective with closed finite-dimensional range. The map $Cp \mapsto D^{1/2}p$ is a contraction exactly when $\|D^{1/2}p\|^2 \leq \|Cp\|^2$ for every p , namely $D \preceq G$. It extends by zero on the orthogonal complement of $\text{ran } C$. Existence of this abstract contraction is therefore the same conjecture. A physical construction of it from specified interactions would be additional structure, whose necessity under RH is not established here.

For a simple control, take $W = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ and $\alpha = 1$. Then $H = 1$, $V = 1/2$, $G = 1$, $D = 3/2$, and $S = -1/2$. The positive shifted reference and the positive high block coexist with negative full-form energy. The conjectural ordering is an essential step.

9 Residual identity, inverse order and Galerkin convergence

For a trial map $Y : PX_L \rightarrow \text{Dom } H$, let $R = B - HY$. Expand without reordering any noncommuting factors:

$$R^*H^{-1}R = B^*H^{-1}B - B^*Y - Y^*B + Y^*HY.$$

This proves

$$B^*H^{-1}B = B^*Y + Y^*B - Y^*HY + R^*H^{-1}R. \quad (9.1)$$

The form inequality $H \geq \Lambda > 0$ reverses for inverses: $H^{-1} \leq \Lambda^{-1}$. One proof uses

$$\langle z, H^{-1}z \rangle = \sup_v \{2 \text{Re} \langle z, v \rangle - \|H^{1/2}v\|^2\} \leq \sup_v \{2 \text{Re} \langle z, v \rangle - \|\Lambda^{1/2}v\|^2\}.$$

The form-domain inclusion makes the supremum comparison valid. Thus $R^*H^{-1}R \leq R^*\Lambda^{-1}R$.

Let $E_K = P_{N,K}$ and solve $E_K H E_K Y_K = E_K B$ in $E_K X_L$. The finite coefficient matrix is at least $\delta_0 I$ and hence invertible. Multiplying by Y_K^* gives

$$Y_K^* H Y_K = Y_K^* B = B^* Y_K, \quad E_K(B - H Y_K) = 0.$$

The residual R_K has no modes below K , so

$$A - B^*Y_K - d_K^{-1}R_K^*R_K \preceq S_L \preceq A - B^*Y_K. \quad (9.2)$$

The lower expression, denoted $L_{L,K}$, includes the entire residual Gram. Omitting it leaves an upper bound.

For the enclosure theorem alone there is a short convergence argument. Write $E = Q(K_L + R_L + \beta_L I)Q$, so $H = \Lambda + E$ with E bounded. Coercivity of the finite systems and Galerkin orthogonality give

$$\|Y_K\| \leq \delta_0^{-1}\|B\|, \quad R_K = (I - P_K)(B - EY_K), \quad \|R_K\| \leq \|B\|(1 + \|E\|/\delta_0).$$

Therefore the enclosure gap is at most $\|B\|^2(1 + \|E\|/\delta_0)^2/d_K$, which tends to zero. For Hermitian ordered matrices $L_{L,K} \preceq S_L \preceq U_{L,K}$, both errors are bounded in norm by the gap. The digamma asymptotic gives the crude rate $O_L(1/\log K)$. This is an analytic error bound with an L -dependent constant, not an estimate of the computational work needed to enclose that constant.

For the stronger convergence $R_K \rightarrow 0$ used in Section 10, each exact solution column $u = H^{-1}Be_j$ minimizes the coercive energy with that source. The Galerkin solution u_K is its orthogonal projection in the H -form inner product onto the trial space. The cosine core gives $\|H^{1/2}(u - u_K)\| \rightarrow 0$; in particular $u_K \rightarrow u$ in L^2 . There are finitely many source columns, so this is operator-norm convergence on the input space. Write $H = \Lambda + E$ with E bounded, using (5.3). Then

$$R_K = (I - P_K)(B - EY_K).$$

Here $(I - P_K)\Lambda Y_K = 0$ and the other residual rows vanish by Galerkin orthogonality. Since $EY_K \rightarrow EH^{-1}B$ in norm and the input space is finite dimensional, $R_K \rightarrow 0$ in norm. Consequently the gap between the two matrices in (9.2) tends to zero.

For ordered inverse approximations the direction of the bound is easy to reverse accidentally. If $F_\ell \preceq H^{-1}$, then $A - B^*F_\ell B \succeq S_L$. For example with $H = 1$, $\alpha = 1$, $V = 1/2$, $F_\ell = \sum_{j=0}^{\ell} 2^{-j-1} = 1 - 2^{-\ell-1}$ and $A = 1 - 2^{-\ell-2}$, one has

$$A - F_\ell = 2^{-\ell-2} > 0, \quad A - H^{-1} = -2^{-\ell-2} < 0.$$

Positive approximate upper matrices cannot certify the target.

10 A fully specified finite-trial error allowance

This section makes the word “approximable” operational. It is a consequence of the boundary column estimate and the residual identity, with an explicit allowance for a finite mass approximation. It is not an executed positivity certificate for any arithmetic matrix.

Choose $K > N$ and any matrix $Y : PX_L \rightarrow P_{N,K}X_L$, for example one with rational coefficients. No exact inverse or exact Galerkin solution is required. Set $V_0 = P - Y$ as a map from the low coefficient space to P_KX_L . The residual identity (9.1) has the variational form

$$C(Y) = V_0^*W_LV_0, \quad R(Y) = QW_LV_0 = B - HY, \quad S_L = C(Y) - R(Y)^*H^{-1}R(Y). \quad (10.1)$$

Thus the energy of an arbitrary trial exceeds the exact minimized energy by its remaining response. Define

$$W^{(J)} = b(H_N) + K^{(J)} + R_L, \quad C_J = V_0^*W^{(J)}V_0, \quad \tilde{R} = QW^{(J)}V_0.$$

The true residual is $R = QW_L V_0 = B - HY$. Positivity of $K_L - K^{(J)}$ gives $V_0^* W_L V_0 \succeq C_J$, while (6.5) gives

$$\|\Lambda^{-1/2}(R - \tilde{R})\| \leq \frac{\eta_{K,J}\|V_0\|}{\sqrt{\delta_0}} =: \epsilon_{K,J,Y}, \quad \|V_0\|^2 = 1 + \|Y\|^2. \quad (10.2)$$

The last equality uses orthogonality of the low and trial high spaces.

For a chosen $M > N$, let $r_j = e_j^* \tilde{R}$ and define the finite matrix

$$T_M = \sum_{j=N}^{M-1} \frac{r_j^* r_j}{d_j} + \frac{1}{d_M} \left(\tilde{R}^* \tilde{R} - \sum_{j=N}^{M-1} r_j^* r_j \right). \quad (10.3)$$

It is positive and bounds $\tilde{R}^* \Lambda^{-1} \tilde{R}$ from above, by Parseval and monotonicity of d_j . All its quantities are finite integrals of the explicitly given functions in (6.8), with $K^{(J)}$ in place of K_L . In particular the full residual norm is kept even if only finitely many cosine coefficients are individually computed.

For operators X and Z with $\|Z\| \leq \epsilon$,

$$(X + Z)^*(X + Z) \preceq X^* X + (2\|X\|\epsilon + \epsilon^2)I.$$

Apply this with $X = \Lambda^{-1/2} \tilde{R}$ and use $\|X\| \leq \sqrt{\|T_M\|}$. Equations (9.1) and (10.2) then prove the rigorous lower bound

$$S_L \succeq C_J - T_M - \left(2\sqrt{\|T_M\|} \epsilon_{K,J,Y} + \epsilon_{K,J,Y}^2 \right) I. \quad (10.4)$$

This uses only finite matrices, integrals of finite sums, and explicit scalar error bounds. Extra numerical rounding or integration uncertainty must be subtracted as an additional scalar operator-norm allowance. If entrywise Hermitian errors are bounded by $e_{ij} \geq 0$, the maximal row sum of the symmetric error majorant bounds their operator norm. One may use rational upper bounds for all norms and lower bounds for the positive d_j and δ_0 .

The special-function values b_j can be enclosed by (3.4). Exponentials, logarithms and trigonometric values in the finite entries admit convergent elementary series with rational remainder estimates. Section 4 supplies a terminating certification procedure for the cutoff and for the active prime powers at integer lengths. It does not require deciding equality in the optional mathematical minimum.

For fixed trial Y and K , let $J \rightarrow \infty$ and refine scalar enclosures; then $\epsilon_{K,J,Y} \rightarrow 0$, $C_J \rightarrow V_0^* W_L V_0$, and the residual approximations converge in L^2 . Exact Galerkin trials with $K \rightarrow \infty$ have vanishing true residual by the preceding section. Approximating their finite coefficients by rationals and taking sufficiently many masses therefore makes (10.4) as close to S_L as needed. This proves availability of finite error control, not a bound on the cost or a sign for the resulting matrix.

11 The tolerance conjecture and its quantifiers

Define $L_{n,K}$ by (9.2). The alternate conjecture is

$$\forall n, m \in \mathbb{N} \quad \exists K > N(n) : \quad L_{n,K} + 2^{-m} I \succ 0. \quad (11.1)$$

If $S_n \succeq 0$, norm convergence gives a K such that $\|L_{n,K} - S_n\| < 2^{-m}$. Thus the displayed matrix is strictly positive. Conversely a witness K gives $S_n \succeq L_{n,K} \succ -2^{-m} I$. Since this holds for every m , each eigenvalue of S_n is nonnegative. Combining with Section 7 proves that (11.1) is equivalent to RH.

The order of quantifiers matters. Neither a single K for all m nor a uniform K or N for all n is asserted. Replacing “every m ” by one small tolerance would be weaker than positivity. Requiring

an unshifted strict finite certificate for each n is stronger on its face and would need a separate strict-positivity argument to claim equivalence. The tolerance formulation avoids this unnecessary extra premise.

The strict margin in (11.1) also permits replacing the exact matrix by a sufficiently accurate certified lower enclosure such as (10.4). Under $S_n \succeq 0$, choose all approximation errors less than a fixed fraction of 2^{-m} . The existence argument then has slack. It remains necessary to prove the arithmetic inequalities for all n, m ; the generic convergence theorem is not that proof.

12 The exact prime discrepancy and a negative control

Let $h_f(r) = \text{Re}\langle F, U_r F \rangle$ and $E(r) = \psi_{\text{Ch}}(e^r) - e^r + 1$. Define

$$t_{\geq 1, L}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{k \geq 1} \frac{2}{a_k} \frac{\tau^2}{a_k^2 + \tau^2} |\widehat{E_L f}(\tau)|^2 d\tau,$$

the gamma form after removal of its $k = 0$ mass. The prime term is $-2 \int_0^L e^{-r/2} h_f(r) d\psi_{\text{Ch}}(e^r)$. The pole term is $4 \int_0^L \cosh(r/2) h_f(r) dr$. Subtracting the smooth prime density $e^r dr$ from this pole expression leaves $2 \int_0^L e^{-r/2} h_f(r) dr$, the form of the kernel $e^{-|x-y|/2}$. The $a_0 = 1/2$ gamma summand is $4I$ minus precisely that kernel, since $2a_0 R_{f, a_0}$ has kernel $e^{-|x-y|/2}$. Thus

$$Q_L[f] = t_{\geq 1, L}[f] + (4 + w_0) \|f\|^2 - 2 \int_0^L e^{-r/2} h_f(r) dE(r). \quad (12.1)$$

Autocorrelations of L^2 functions are continuous, and the finite-interval measure has finite variation. This is a bounded form contribution, so the identity extends from the smooth core to $\mathcal{D}_L$.

For smooth compact support, $E(0) = 0$ and $h_f(L) = 0$ remove both boundary terms in Stieltjes integration by parts. The last term becomes $2 \int_0^L E(r) e^{-r/2} (h'_f(r) - h_f(r)/2) dr$. No such derivative is assumed to be an ordinary function on the whole logarithmic domain.

If the prime measure is replaced by its leading density, then $E = 0$. For a zero extension in $H^1(\mathbb{R})$,

$$t_{\geq 1, L}[f] \leq \sum_{k \geq 1} \frac{2}{a_k^3} \|F'\|^2 \leq \frac{26}{125} \|F'\|^2.$$

The last fraction is $2/(5/2)^3 + 1/[2(5/2)^2]$ by an integral tail. Equation (2.3) implies $w_0 < -5$. To check the elementary inequality, use $\gamma_E > 1/2$, $\pi > 3$, $\log 2 > 2/3$ and $\log \pi > 1$. For example $\gamma_E > H_8 - \log 9 > H_8 - 11/5 = 29/56 > 1/2$; $e^{11/10} > 3$ follows from its first six positive series terms, and $e < 3$ gives $\log \pi > 1$.

On $(0, 2)$, $f(x) = \sin(\pi x/2)$ has norm one, vanishes at the endpoints, and $\|F'\|^2 = \pi^2/4$. Hence

$$Q_2^{\text{dens}}[f] < -1 + \frac{26}{125} \frac{(22/7)^2}{4} = -\frac{2979}{6125}.$$

Approximation in $H_0^1(0, 2)$ preserves negativity and gives smooth compact support. This shows why simply discarding the signed prime discrepancy does not provide a route to the matrix ordering. It does not rule out methods that use stronger, signed prime information.

13 The length-one high-mode example

At $L = 1$ only $p = 2, m = 1$ is active. The chain has two sites, so

$$\beta_1 = -w_0 + 2 \sinh(1/2) - 1 + \frac{\log 2}{\sqrt{2}}.$$

The longer manuscript supplies the rational bounds

$$\begin{aligned} \gamma_E &< 0.58, \quad \pi < 22/7, \quad \log 2 < 0.7, \\ \log \pi &< 23/20, \quad 2 \sinh(1/2) - 1 < 43/1000, \quad \log 2/\sqrt{2} < 1/2. \end{aligned}$$

They give

$$\beta_1 < \frac{58}{100} + \frac{11}{7} + \frac{21}{10} + \frac{23}{20} + \frac{43}{1000} + \frac{1}{2} = \frac{41611}{7000}.$$

These constants may be checked using rational series: the atanh series $\log 2 = 2 \sum_{j \geq 0} 3^{-(2j+1)}/(2j+1)$ gives its upper and lower bounds; $\gamma_E < H_{256} - \log 256$ gives the displayed Euler-constant bound with $\log 2 > 6931/10000$; positive exponential sums and geometric tail bounds give the remaining logarithm and sinh bounds. For $\pi < 22/7$ use $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2)dx > 0$.

Since $\pi > 3$ and b is increasing,

$$b((4\pi)^2) > \sum_{k=0}^{63} \frac{2}{a_k} \frac{144}{a_k^2 + 144} > \frac{3007}{500} > \frac{41611}{7000} + \frac{1}{16}.$$

The two final comparisons use exact fractions. Thus $N = 4$ is an admissible cutoff with a strict margin. It proves $N_{\min}(1) \leq 4$ for the optional mathematical minimum, and permits a deliberate four-dimensional low space. It does not specify the output of the stagewise search for an arbitrary fixed enclosure routine. No arithmetic Schur-matrix entries or sign certificate are inferred from these scalar inequalities.

14 Human review record

Mark an item complete only after checking its proof and any cited premise. This list starts unreviewed. A mathematical question should be recorded with the equation number and the needed correction, then resolved before that item is marked complete.

Required check	Reviewer / date
Weil normalization and Fourier bridge, Sections 2–3	_____
Closed logarithmic form and smooth core, Section 3	_____
Neumann comparison and certified cutoff, Section 4	_____
Green-kernel sign and equality of domains, Section 5	_____
Carleman essential norm and finite-input tails, Sections 5–6	_____
Prime/pole entries, parity and both tails, Section 6	_____
Schur identity and both RH implications, Section 7	_____
Shift cancellation and projected source, Section 8	_____
Residual bounds and both convergence proofs, Section 9	_____
Finite-trial error propagation, Section 10	_____
Tolerance quantifiers and null directions, Section 11	_____
Discrepancy sign and rational control, Sections 12–13	_____
Reference accuracy, prior work and claim of novelty	_____
Final text and disclosure match the review actually done	_____

Completing these items validates the proposed reduction only to the extent of the actual human checks; it does not prove the open conjecture and does not constitute independent peer review. The original package’s 59 algebra checks reproduce, but they do not certify the analytic arguments above. The physical portions omitted from the short paper need their own review if they are later submitted as separate mathematical claims.

Preparation note

OpenAI’s Codex assisted substantially with the analysis, calculations, code and writing of this companion. The finite tolerance formulation and the explicit finite-trial allowance are developed here from the residual and column estimates in the source investigation. They are supplied with proofs for review, not represented as externally checked results. This revision also incorporates the documented Codex review of 13 September 2026, including the certified-cutoff convention, normalization bridge, projected source, parity decomposition and sharper compressed tail. No human review has yet been recorded in this file.

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- [6] *Arithmetic ground-state geometry: Physical caps, arithmetic sources and analytical sign tests*, working manuscript in the [shifted-zeta-positivity](#) project, September 2026, [repository revision efb908a](#), especially Sections 10, 14 and 15.